



Auckland Knights Chess Club

Hosting and Deployment Guide

Prepared for Auckland Knights Chess Club

Version 1.0 | 24 May 2026

This document is part of the Auckland Knights Chess Club product documentation pack.

Confidential working documentation for club setup, deployment and administration.

Table of Contents

1. Purpose and system overview
2. Recommended deployment stack
3. Accounts and prerequisites
4. Local project preparation
5. Supabase setup
6. Stripe setup
7. Email provider setup
8. Address autocomplete setup
9. Vercel deployment
10. Post-deployment configuration
11. Production smoke test
12. Operations, security and troubleshooting
13. References

1. Purpose and System Overview

This guide explains how to deploy the Auckland Knights Chess Club web application to production using a low-cost, scalable stack suitable for a new club. It is written for a technical owner, developer, or trusted club administrator.

The current application is a Next.js web application with public pages, a hidden admin area, Supabase database/auth/storage, Stripe payments, optional email providers, and PWA install support.

Item	Value / Guidance
Application name	Auckland Knights Chess Club Web Application
Latest feature baseline	v4.x product build, including PWA support, social share assistant, calendar enrolment, membership ID, reports and email provider selection
Primary hosting recommendation	Vercel Free/Hobby initially, upgrade later only when usage requires it
Backend recommendation	Supabase Free initially, then Supabase Pro when official production reliability/backups are needed
Payments	Stripe Checkout and webhook confirmation
Email	Brevo SMTP or Resend API, selected in Club Settings

Important: Never commit .env.local, Supabase service role keys, Stripe secret keys, webhook secrets, SMTP passwords, or Resend API keys to GitHub.

2. Recommended Deployment Stack

Layer	Recommended service	Purpose
Frontend hosting	Vercel	Hosts the Next.js web application, public pages, admin pages and server routes.
Database/Auth/Storage	Supabase	Stores members, tournaments, news, results, files, admin profiles and authentication.
Payments	Stripe	Handles membership and general/open tournament payments.
Email	Brevo SMTP or Resend API	Sends bulk email, notices, absence notifications, contact emails and confirmations.
Address autocomplete	Addy or fallback provider	Populates New Zealand addresses where address fields are used.
Code repository	GitHub	Source control and Vercel deployment source.

Vercel is recommended because it is purpose-built for Next.js deployments and supports environment variables, automatic deployments from GitHub, HTTPS and custom domains. Supabase provides a managed PostgreSQL database, authentication and storage in one place. Stripe Checkout provides hosted payment pages and uses webhooks to confirm completion.

3. Accounts and Prerequisites

- GitHub account for the source repository.
- Vercel account connected to GitHub.
- Supabase account and project.

- Stripe account in Test Mode first, then Live Mode when ready.
- Brevo account or Resend account for emails.
- Optional domain such as aucklandknights.co.nz.
- Node.js LTS installed locally.
- A working local copy of the latest project ZIP.

Recommended local command check:

```
node -v
npm -v
```

4. Local Project Preparation

Unzip the latest project into a clean folder. Use the public npm registry to avoid internal registry lock-file problems.

```
cd C:\auckland-knights-black-chess-v4-3-pwa-social-share
npm config set registry https://registry.npmjs.org/
if exist package-lock.json del package-lock.json
if exist node_modules rmdir /S /Q node_modules
npm install --registry=https://registry.npmjs.org/
npm run dev
```

Open the local site at <http://localhost:3000> and admin login at <http://localhost:3000/club-admin/login>.

If a folder delete fails with EPERM, close VS Code, browsers and terminals, then retry or rename node_modules to node_modules_old and reinstall.

5. Supabase Setup

Create a new Supabase project for production. Use the Supabase SQL Editor to run the schema and migration files provided in the project ZIP.

For a fresh production database:

```
supabase/schema.sql
# Then run any later migration files in version order if supplied.
```

For an existing test database refresh:

```
supabase/reset_database.sql
supabase/schema.sql
# Then run migration files in order.
```

Do not directly delete from Supabase storage.objects or storage.buckets using SQL. Delete uploaded files from the Supabase Storage UI or Storage API.

Required storage buckets vary by version, but generally include:

- news-images
- gallery-images
- tournament-images
- coach-images
- agm-attachments

- pwa-icons if used

Create the first Super Admin:

```
-- 1. Create the user in Supabase Authentication > Users first.
-- 2. Then run this SQL, replacing the email and names.
insert into public.profiles (id, email, first_name, last_name, full_name, role, is_active, created_at,
updated_at)
select u.id, u.email, 'Bharat', 'Somaraju', 'Bharat Somaraju', 'super_admin', true, now(), now()
from auth.users u
where lower(u.email) = lower('your-email@example.com')
on conflict (id) do update set
role = 'super_admin',
is_active = true,
updated_at = now();
```

6. Stripe Setup

Start in Stripe Test Mode. The application creates Stripe Checkout Sessions and marks membership/general tournament payments confirmed only after the webhook receives a successful checkout.session.completed event.

- Copy STRIPE_SECRET_KEY from Stripe Developers > API keys. It should start with sk_test_ for testing or sk_live_ for production.
- Create a webhook endpoint for the deployed URL: <https://your-domain/api/stripe/webhook>.
- Enable event checkout.session.completed.
- Copy the webhook signing secret that starts with whsec_ into STRIPE_WEBHOOK_SECRET.
- Redeploy after changing Vercel environment variables.

Local webhook testing:

```
stripe login
stripe listen --forward-to localhost:3000/api/stripe/webhook
```

7. Email Provider Setup

The product supports email provider selection from Club Settings. Use either Brevo SMTP, custom SMTP, Resend API or Disabled.

Provider	Use case	Typical settings
Brevo SMTP	Recommended practical free/low-cost option for club notices and bulk email.	Host: smtp-relay.brevo.com, Port: 587, Secure: false, User: SMTP login, Password: SMTP key.
Resend API	Simple API-based sending for transactional emails.	RESEND_API_KEY and verified sender/domain.
Disabled	For testing when no email should be sent.	Email templates and drafts remain usable but sending is disabled.

Brevo SMTP environment example:

```
EMAIL_PROVIDER=smtp
SMTP_HOST=smtp-relay.brevo.com
SMTP_PORT=587
SMTP_SECURE=false
SMTP_USER=your_brevo_smtp_login
SMTP_PASS=your_brevo_smtp_key
CLUB_FROM_EMAIL=verified_sender_email
CLUB_FROM_NAME=Auckland Knights Chess Club
CLUB_REPLY_TO_EMAIL=info@aucklandknights.co.nz
```

Use Email Diagnostics in the admin area to test connection, send a test email and inspect provider status.

8. Address Autocomplete Setup

The application can use Addy for New Zealand address autocomplete and can fall back to a basic provider. The recommended provider is Addy for better NZ address results.

```
ADDRESS_PROVIDER=addy
ADDY_API_KEY=your_addy_key
ADDY_API_URL=
```

Address autocomplete should be available in club settings address, tournament venue address, membership address, calendar venue and other address fields.

9. Vercel Deployment

Push the project to GitHub first.

```
git init
git add .
git commit -m "Initial Auckland Knights deployment"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/REPOSITORY.git
git push -u origin main
```

In Vercel: Add New Project > Import GitHub repository > Framework: Next.js > Deploy.

Add these environment variables in Vercel Project Settings > Environment Variables:

```
NEXT_PUBLIC_SITE_URL=https://your-domain-or-vercel-url
NEXT_PUBLIC_SUPABASE_URL=https://xxxxx.supabase.co
NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY=...
NEXT_PUBLIC_SUPABASE_ANON_KEY=...
SUPABASE_SERVICE_ROLE_KEY=...
STRIPE_SECRET_KEY=sk_live_or_sk_test...
STRIPE_WEBHOOK_SECRET=whsec...
EMAIL_PROVIDER=smtp
SMTP_HOST=smtp-relay.brevo.com
SMTP_PORT=587
SMTP_SECURE=false
SMTP_USER=...
SMTP_PASS=...
CLUB_FROM_EMAIL=...
CLUB_FROM_NAME=Auckland Knights Chess Club
CLUB_REPLY_TO_EMAIL=...
ADDRESS_PROVIDER=addy
ADDY_API_KEY=...
```

After changing Vercel environment variables, redeploy the application so the new values are used.

10. Post-deployment Configuration

After the first Vercel deployment, configure external systems with the real domain or Vercel deployment URL.

System	Configuration
Supabase Auth URLs	Set Site URL to the production URL. Add redirect URLs for production and localhost testing.
Stripe webhook	Endpoint: <code>https://your-domain/api/stripe/webhook</code> , event: <code>checkout.session.completed</code> .
Resend/Brevo sender	Verify sender email/domain before production email sending.
Vercel domain	Add custom domain under Vercel Project > Settings > Domains.
Club Settings	Publish settings for contact details, social links, email provider, membership prefix and calendar title.

11. Production Smoke Test

Run this checklist after deployment and before giving the site to club users.

Area	Test
Public pages	Home, Calendar, Tournaments, Results, News, Gallery, Coaching, Contact, Install App.
Admin login	Login at <code>/club-admin/login</code> as Super Admin.
Club settings	Update and publish contact details, social links, membership prefix and email provider.
Membership	Submit membership registration and complete Stripe test payment. Check Membership ID generation.
Club calendar event enrolment	Enter active Membership ID or email, fetch member details, confirm enrolment without payment.
General/open tournament	Register for paid event and confirm only after Stripe webhook.
Email	Use Email Diagnostics and send test email.
Reports	Preview table, show PDF preview, export CSV/PDF.
PWA	Open <code>/install-app</code> and test Add to Home Screen instructions.

12. Operations, Security and Troubleshooting

Operational recommendations:

- Keep a Super Admin account for Bharat/club owner and a second backup Super Admin only if needed.
- Use least privilege roles for volunteers: `news_editor`, `tournament_manager`, `gallery_manager`, `social_media_manager`, etc.
- Store secrets only in Vercel environment variables or local `.env.local`, never in GitHub.
- Use Stripe Test Mode until registration/payment flow is fully verified.
- Keep Supabase backups/export strategy before production usage grows.

- Rotate keys if they were ever shared in screenshots or chats.
- Keep package-lock.json regenerated on the public npm registry if npm tries to use an internal registry.

Common fixes:

Problem	Fix
npm tries packages.applied-caas...	Delete package-lock.json, set npm registry to https://registry.npmjs.org/ , reinstall.
Brevo unauthorized IP	Authorise the IP in Brevo, use SMTP login and SMTP key, not account password.
Payments stay pending	Check Stripe webhook endpoint and STRIPE_WEBHOOK_SECRET.
Admin has no access	Confirm profiles.role = super_admin/admin and is_active = true.
Images not displaying	Check storage bucket is public or correct signed/public URL is saved.

13. References

Official sources used for deployment practices and provider-specific configuration:

- Vercel documentation - Environment variables and deployments: <https://vercel.com/docs/environment-variables>
- Vercel documentation - GitHub deployments: <https://vercel.com/docs/git/vercel-for-github>
- Supabase documentation - Authentication redirect URLs: <https://supabase.com/docs/guides/auth/redirect-urls>
- Stripe documentation - Webhooks: <https://docs.stripe.com/webhooks>
- Stripe documentation - Checkout session completed event: <https://docs.stripe.com/api/events/types>
- Brevo documentation - SMTP integration: <https://developers.brevo.com/docs/smtp-integration>
- Brevo documentation - SMTP ports: <https://help.brevo.com/hc/en-us/articles/10905415650322-Which-SMT-P-port-should-I-use-Port-587-465-or-2525>